

Javon Darby
Prof. Anna Devarakonda
ITAI 2372
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AI in Retail

Overview

- AI is transforming the retail industry
- Focus areas:
 - Customer experience
 - Store operations
 - Supply chain

Retail Technology Evolution

- Retail has evolved due to:
 - Increased computing power
 - More available data
 - Changing customer expectations

Key Drivers of AI in Retail

Customer Expectations

- Personalization
- Convenience
- Speed

Business Pressures

- Competition
- Profit margins
- Need for efficiency

Core AI Technologies

- Machine Learning
- Computer Vision
- Natural Language Processing (NLP)
- Robotics
- Predictive Analytics

Computer Vision in Retail

- Shelf monitoring
- Customer tracking
- Self-checkout systems
- Loss prevention
- Visual search

Natural Language Processing (NLP)

- Chatbots (customer service)
- Voice shopping
- Social media monitoring
- Customer feedback analysis
- Product search optimization

Personalization

- AI recommends products based on:
 - Customer behavior
 - Preferences
 - Purchase history

Virtual & Augmented Reality

- Virtual try-on
- Store navigation
- Product visualization
- Interactive displays
- Employee training

Smart Store Technologies

- Smart shelves
- Interactive mirrors
- Electronic shelf labels
- Automated checkout
- Traffic analysis

Supply Chain AI

- Demand forecasting
- Inventory optimization
- Route optimization
- Supplier management
- Risk prediction

Warehouse Automation

- Picking robots
- Automated guided vehicles (AGVs)
- Sorting systems
- Inventory drones
- Smart storage

AI in Store Operations

- Staff scheduling
- Maintenance prediction
- Energy optimization
- Security systems
- Cleaning automation

Implementation Challenges

- Poor data quality
- Integration issues
- Change management
- Measuring ROI
- Skill gaps

Success Metrics (KPIs)

- Customer satisfaction
- Operational efficiency
- Revenue growth
- Cost reduction
- Employee adoption

Ethical Considerations

- Privacy
- Transparency
- Bias in AI systems

Key Takeaways

- AI is changing all areas of retail
- Businesses must adapt to stay competitive
- AI improves efficiency and customer experience
- Proper planning and ethics are important